An Introduction to Automatic Speech Recognition (ASR)

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Artificial Intelligence for Customer Service

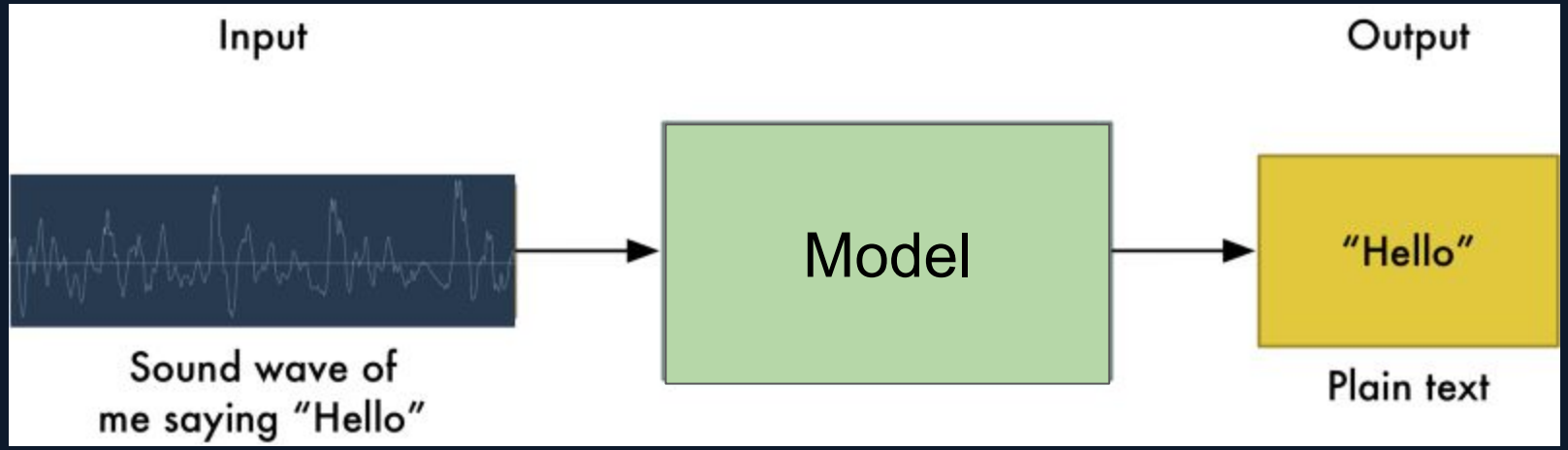
Outline

- Introduction
- Classical ASR Pipeline
  - Input Audio Signal
  - Signal Processing and Feature Extraction
  - Acoustic Model
  - Hypothesis Search
  - Language Model
- Evaluation Metric
- End to End Approach
  - LAS - Listen Attend Spell

Introduction

What is ASR?

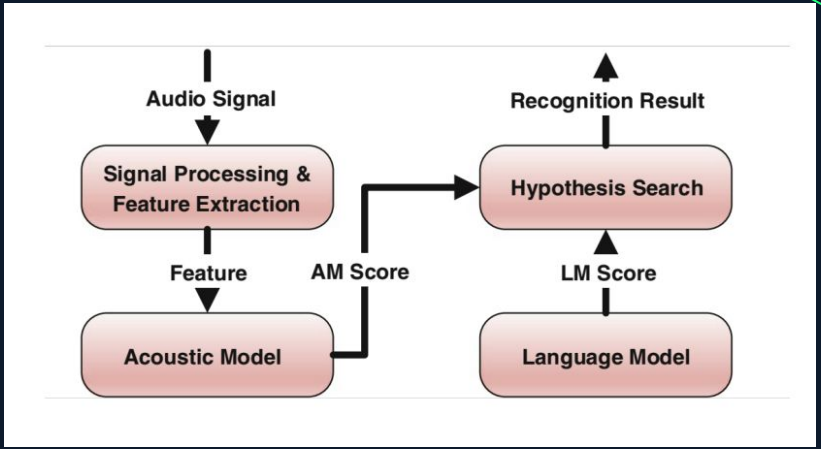
It is the task of detecting spoken words from an audio signal.



Source - https://medium.com/@ageitgey/machine-learning-is-fun-part-6-how-to-do-speech-recognition-with-deep-learning-28293c162f7a

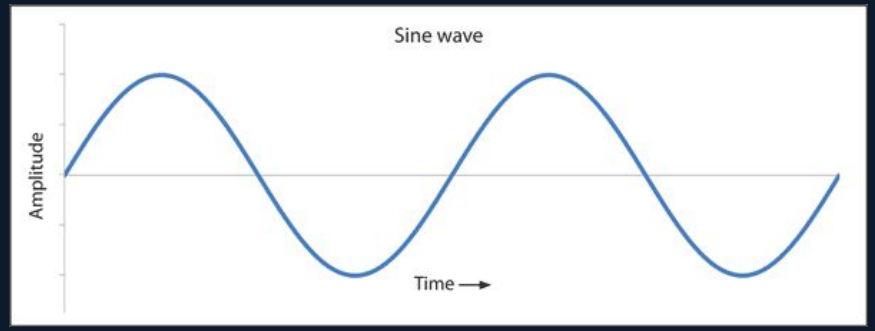
Classical ASR Pipeline

1. **Signal processing and feature extraction** - cleans the audio signal and extracts feature vectors.
2. **Acoustic model (AM)** - estimates the probability of a hypothesized audio sequence.
3. **Language model (LM)** - estimates the probability of a hypothesized word/character sequence.
4. **Hypothesis search** - combines AM and LM scores and outputs the word sequence with the highest score.



Source - http://125.234.102.146:8080/dspace/bitstream/DNULIB_52011/6853/1/automatic_speech_recognition_a_deep_learning_approach.pdf

What is Sound?



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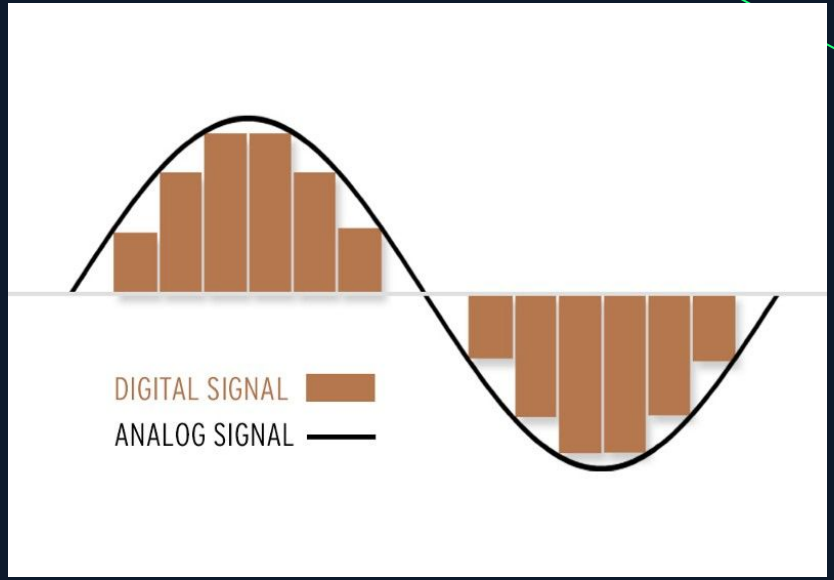
Analog to Digital Signal

- Audio is a continuous analog signal.
- **Sampling Rate** - number of samples collected per second.
- Some common sampling rates are:
  - 44.1KHz
  - 16KHz
- Audio files such as .wav files contains discrete digital signals.
- Use python package — librosa for reading .wav files.



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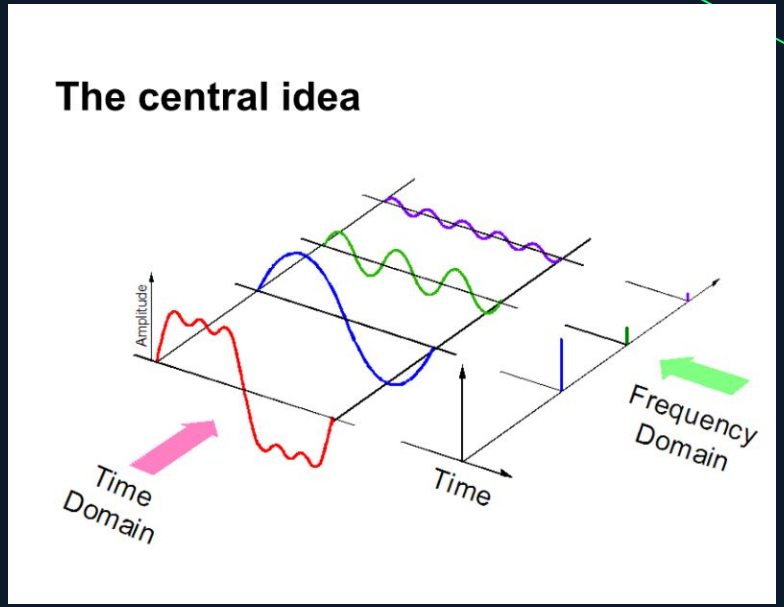
Signal Processing and Feature Extraction

- **Chunking** - Grouping our sampled audio into 20-millisecond-long chunks.
- So now the audio is a sequence of 20-millisecond-long chunks or audio frames.
- **Fourier Series** - Is a way to represent a function as the sum of simple sine/cosine waves.

Fourier Transform

- **Fourier Series** - Is a way to represent a function as the sum of simple sine/cosine waves.
- **Fourier Transform** - It uses Fourier Series to convert our time-vs-amplitude signal to time-vs-frequency signal.
  - Requires less storage space.
  - Sort of a frequency fingerprint for audio snippet.
- **Spectrogram** - If we repeat this process on every 20 millisecond chunk of audio, we end up with a spectrogram.



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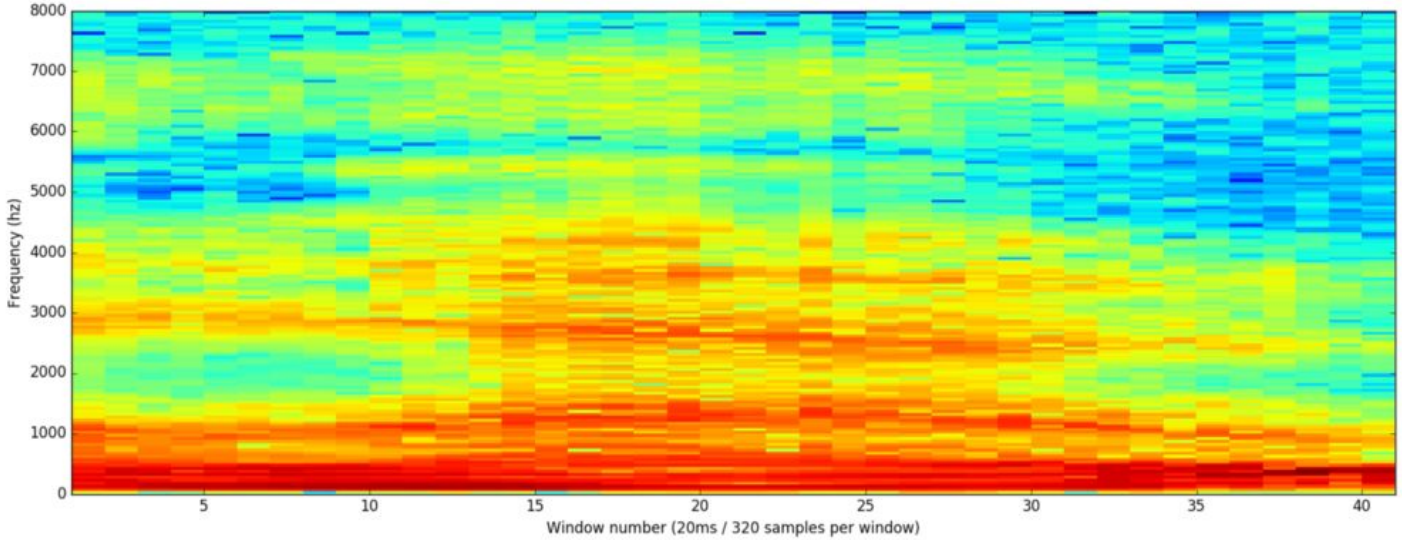
Spectrogram



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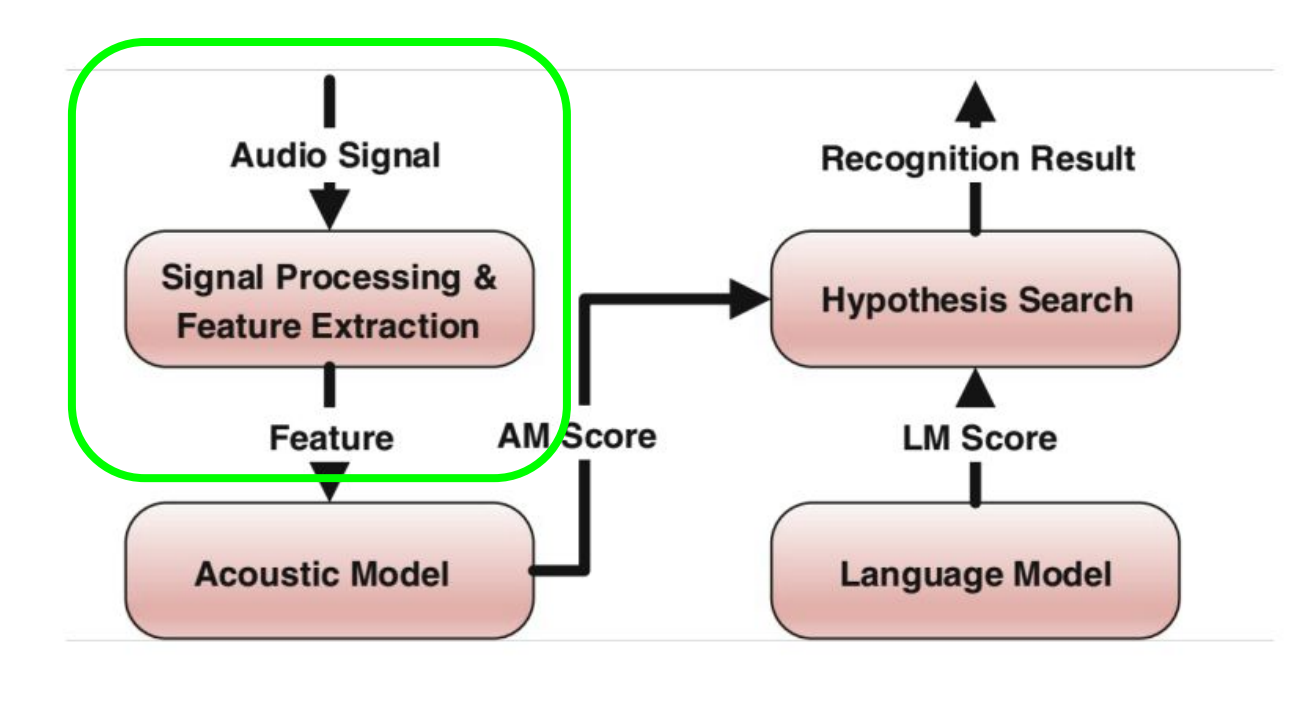
Recap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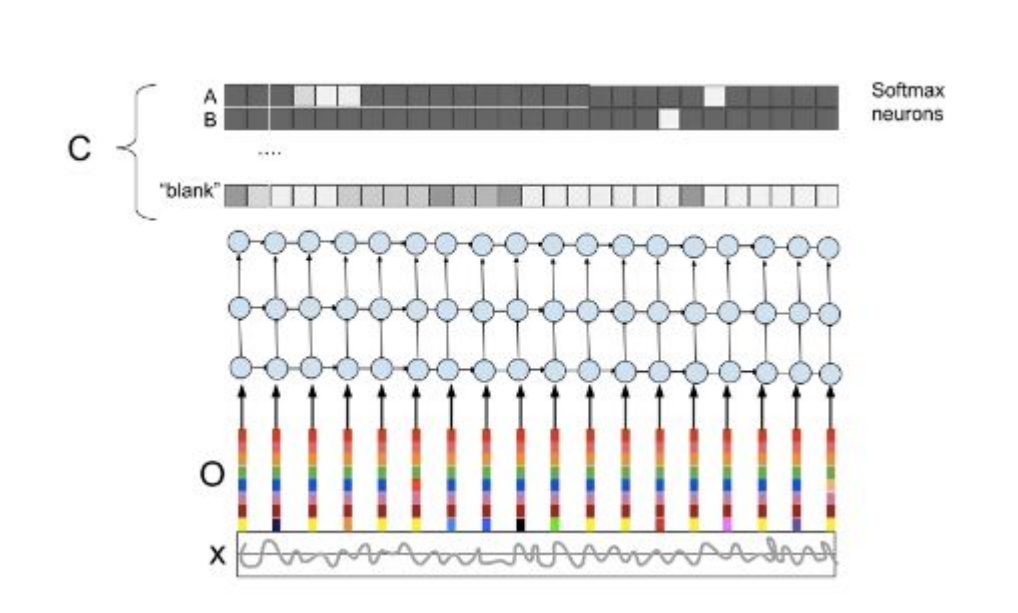
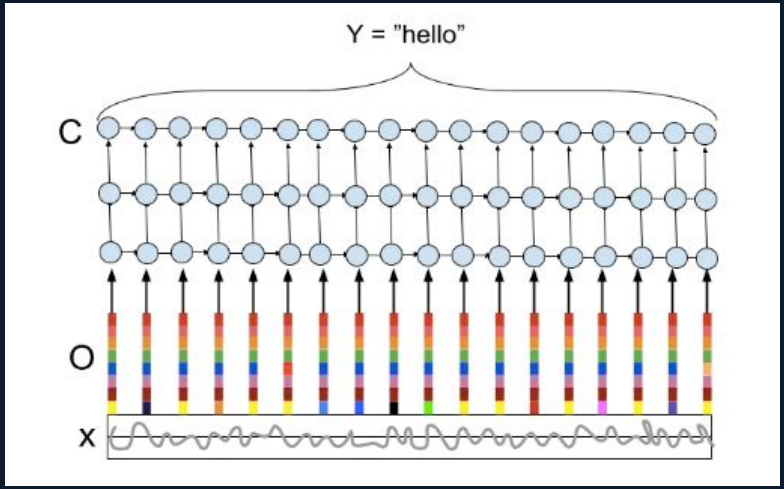
Acoustic Model



Source - https://androidkt.com////recognition-encoding/

Acoustic Model - Challenges

- Length of the input sequence is not the same as the length of the transcription sequence.
  - Use a dataset where labels are chunkwise. Each chunk is mapped either to valid character from the transcript or to an empty character.
    - Labelling is time consuming (boring) and error prone.
    - A single character spanning multiple chunks.
  - Connectionist Temporal Classification (CTC).



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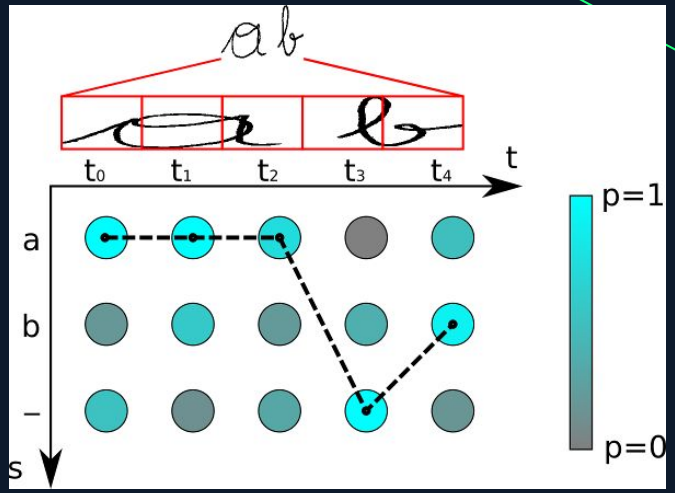
Connectionist Temporal Classification (CTC)

It is a loss that solves both problems for us. With CTC

- we only have to tell the CTC loss function the transcribed ground truth that occurs in the audio. Therefore we ignore both the position and length of the audio and transcription.
- No need to worry about single character spanning over multiple windows as CTC performs a squeezing operation.
  - AAA-B = AB



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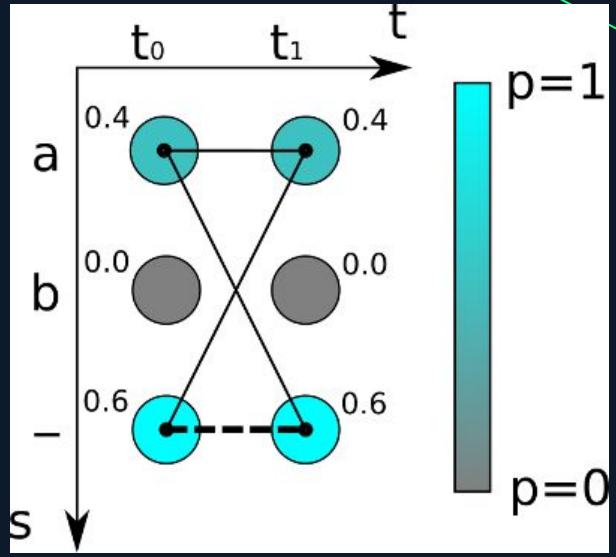
Connectionist Temporal Classification (CTC)

How CTC works?

1. Compute all the possible paths to the ground truth transcription.
   a. AA = A
   b. A- = A
   c. -A = A
2. Compute the scores for these paths by multiplying the corresponding character scores together.
   a. AA = 0.4*0.4 = 0.16
   b. A- = 0.4*0.6 = 0.24
   c. -A = 0.6*0.4 = 0.24



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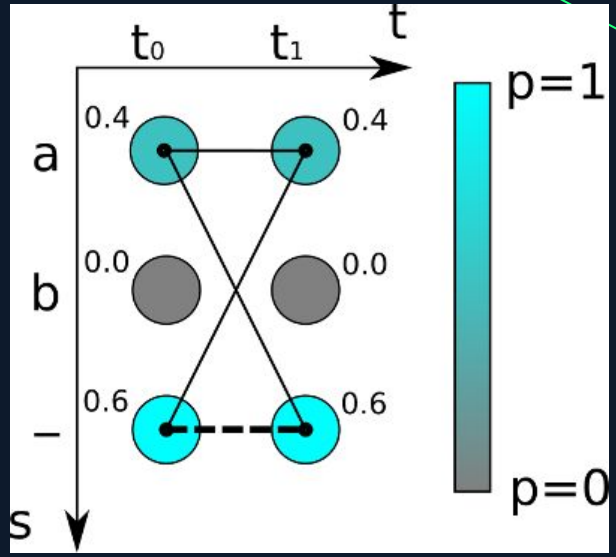
Connectionist Temporal Classification (CTC)

How CTC works?

3. Calculate the score for a given ground truth transcription by summing over the scores of all paths corresponding to this transcription.
   a. A = 0.16 + 0.24 + 0.24 = 0.64
   b. This is the likelihood probability of the ground truth.
4. To compute the loss just take the negative log of the likelihood probability computed in the previous step.



Tensorflow provides the implementation of CTC loss.

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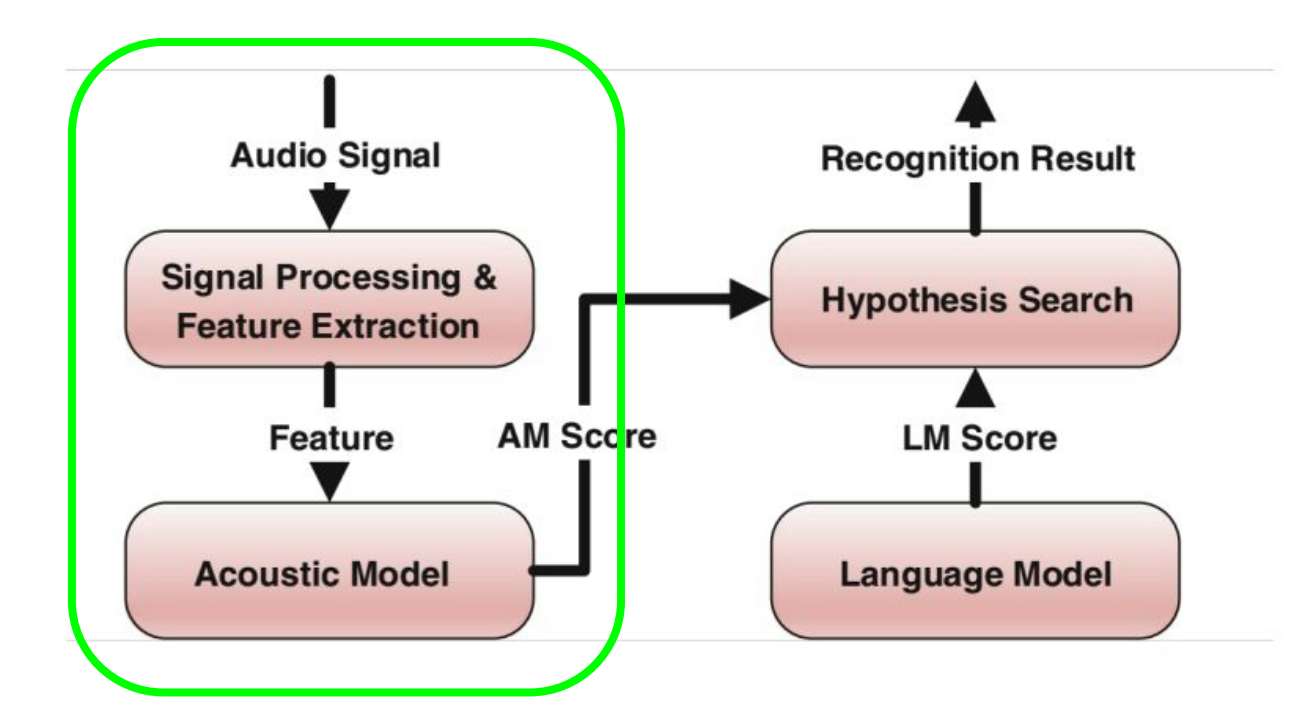
Recap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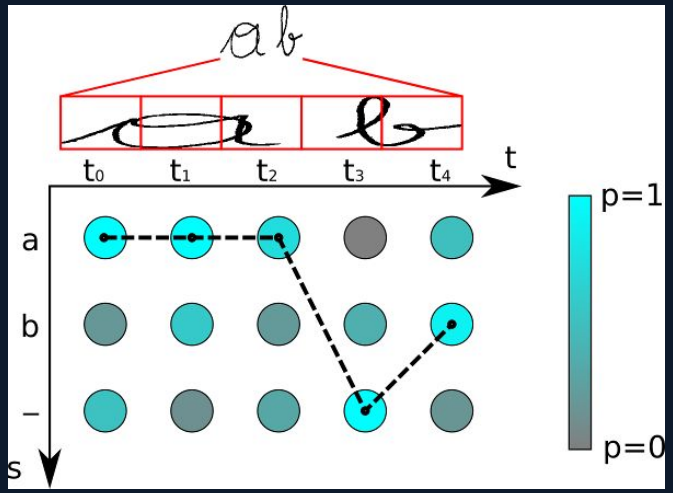
Inference & Hypothesis Search

How to use the Acoustic Model for Inference?

A simple and very fast algorithm is **max decoding** which consists of two steps:

1. It calculates the best path by taking the most likely character per time-step.
2. It then performs a squeeze operation on the encoding by first removing duplicate characters and then removing all blanks from the path. What remains represents the recognized transcription.



Language Model

RNN output	Decoded Transcription
what is the weather like in bostin right now	what is the weather like in boston right now
prime miniter nerenr modi	prime minister narendra modi
arther n tickets for the game	are there any tickets for the game

Table 1: Examples of transcriptions directly from the RNN (left) with errors that are fixed by addition of a language model (right).

- Even with better decoding, CTC models tend to make spelling + linguistic errors.
  - Independent assumption.
  - Not enough audio data to learn complicated spelling and grammatical structure.
  - Very small vocabulary.

Source - https://arxiv.org/pdf/1412.5567.pdf

Language Model

A possible solution is to train a language model on massive text corpora.

- Learn spelling + grammar
- Greatly expanded vocabulary
- Contextual bias

$$Q(c) = \log(\mathbb{P}(c|x)) + \alpha \log(\mathbb{P}_{\text{lm}}(c)) + \beta \text{ word_count}(c)$$

The aim here is to find a sequence c that maximizes the combined objective Q(c).

Source - https://arxiv.org/pdf/1412.5567.pdf

Evaluation Metric - Word Error Rate (WER)

$$WER = \frac{S + D + I}{N}$$

Reference Audio : What a bright day

- **S (Substitutions)** : Number of words replaced.
  - What a light day
- **I (Insertions)** : Number of words that got inserted but were not said.
  - What a beautiful bright day
- **D (Deletions)** : Number of words that got omitted but were said.
  - What a day
- **N** : Total number of spoken words in the reference audio.

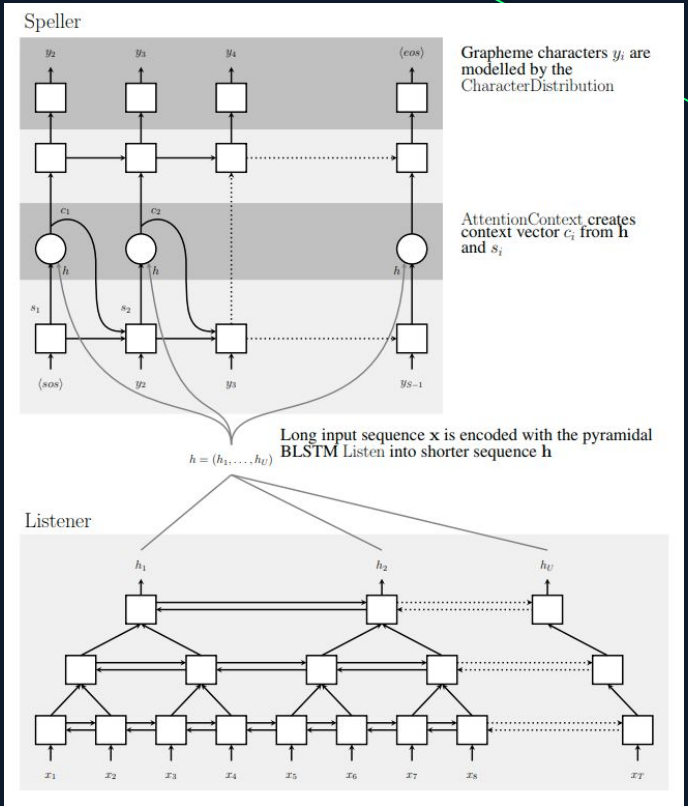
Cons of Pipeline Approach

Cons of pipeline approach for ASR

- A lot of manual effort is required in designing these pipelines.
  - Audio and transcription alignment with CTC
  - Hypothesis search using prefix beam search
- Each component of the pipeline has to be individually trained.
  - Acoustic Model
  - Language Model

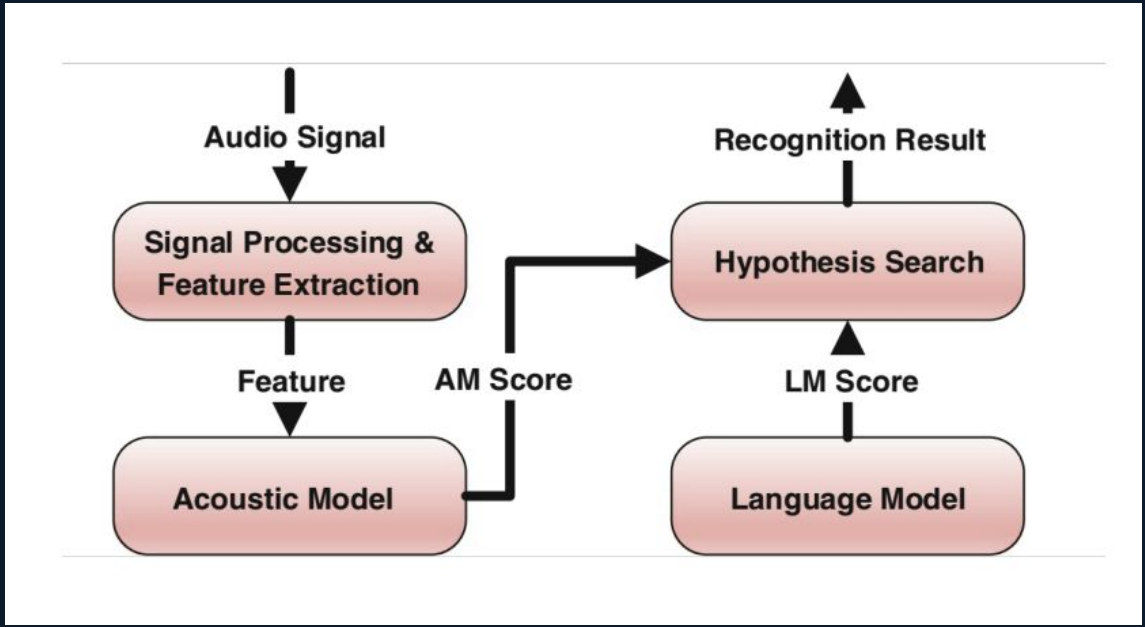
Listen, Attend and Spell (LAS)

- The model uses a Seq-Seq Encoder-Decoder approach.
- Encoder (Listener) is a Pyramidal BiLSTM network.
- Decoder (Speller) is an attention based LSTM network.
- 40000 hours of data.
- A month of training.
- WER of 14% (without language model)
- WER of 10% (with language model)



Source - https://arxiv.org/pdf/1508.01211.pdf

Summary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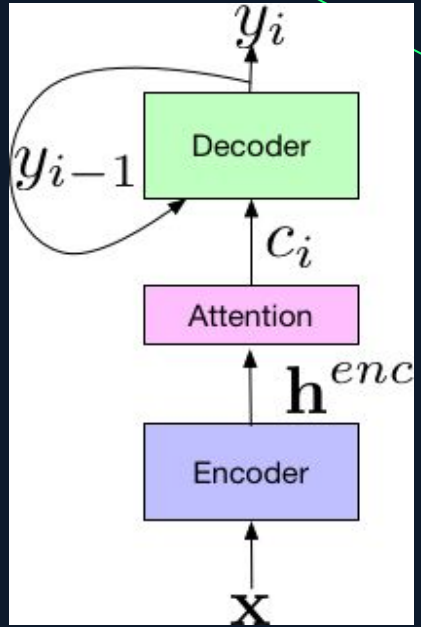
y_i
y_{i-1}
Decoder
c_i
Attention
$\mathbf{h}^{enc}$
Encoder
$\mathbf{x}$

Thank You!

Questions?